PILOT PHASE · SIMULATED KDHS DATA · N = 1,000

Predicting Maternal Health Risk Among Pregnant Women in Kenya Using Machine Learning & Bayesian Inference on the KDHS 2022

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90.0%	0.9717	87%	0	1.003_R	8
Final Ensemble Test Accuracy	Stacking Ensemble AUC-ROC	High-Risk Recall (Critical Class)	High Risk → Low Risk Misclassifications	Bayesian MCMC Convergence (R-hat)	Pipeline Notebooks Completed

📋 BACKGROUND & PROBLEM STATEMENT

Kenya's maternal mortality ratio stands at **355 per 100,000 live births** — five times the SDG 3.1 target — with the majority of preventable deaths concentrated among women who were never screened for risk factors during antenatal care.

Existing tools either operate at the population or county level, or are confined to hospital systems. No individual-level, Bayesian-augmented ML tool exists for use by Community Health Workers (CHWs) in Kenya using nationally representative KDHS data.

> **Research Gap:** CHWs lack a data-driven, probabilistic triage tool calibrated to Kenyan demographic and geographic risk factors.

Dataset: Kenya Demographic & Health Survey 2022 Individual Recode (KDHS 2022-IR). After filtering for women with a recent birth: **10,391 records × 13 variables**. Pilot testing uses 1,000 simulated records mirroring these distributions.

🎯 RESEARCH OBJECTIVES

1. **SO1 — ML Classification:** Develop a TabNet / TabTransformer classification model to stratify pregnant women into Low, Moderate, and High risk categories using KDHS 2022 variables.
2. **SO2 — Bayesian Inference:** Apply PyMC-based Bayesian inference to quantify prediction uncertainty and produce 95% highest-density credible intervals per patient.
3. **SO3 — AI-Integrated App:** Deploy a bilingual (English / Kiswahili) Streamlit application with AI-generated clinical explanations for CHW use.

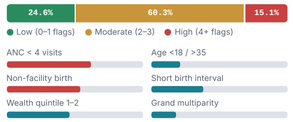
📊 KEY VARIABLES (KDHS 2022)

VARIABLE	DESCRIPTION	RISK FLAG
v012	Mother's age	<18 or >35
v201	Total children ever born	>4
m14_1	ANC visits	<4
m15_1	Delivery location	Non-facility
m17_1	C-section delivery	Yes
b11_01	Birth interval	<24 months
v190	Wealth index	Quintile 1–2
v212	Age at first birth	<18

+ Province (v024), Residence (v025), Education (v106), Marital status (v501)

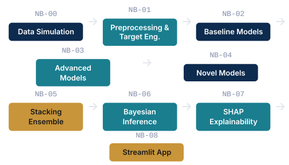
⚙️ TARGET VARIABLE ENGINEERING

Eight WHO-based binary risk flags are summed to a composite score. Thresholds calibrated from pilot distributions:



SMOTE applied on training split only: 700 → 1,269 balanced records.

🔄 PILOT TEST PIPELINE (8 NOTEBOOKS)



📐 NB-02 · TIER 1 BASELINE MODELS

MODEL	ACC.	AUC-ROC	H-RECALL	F1
Logistic Regression	0.660	0.8599	0.783	0.651
Decision Tree	0.640	0.8472	0.739	0.646

Benchmarks set: AUC-ROC > 0.860 · High Recall > 0.783

⚡ NB-03 · TIER 2 ADVANCED MODELS

MODEL	ACC.	AUC-ROC	H-RECALL	F1
Random Forest	0.833	0.944	0.609	0.791
XGBoost	**0.860**	**0.964**	0.609	**0.817**
CatBoost ✦ Champion	0.847	**0.969**	**0.826**	**0.827**

CatBoost selected as Tier 2 champion — only model exceeding baseline High Recall benchmark (0.783).

🧠 NB-04 · NOVEL DEEP LEARNING MODELS

MODEL	ACC.	AUC-ROC	H-RECALL	F1
TabNet	0.687	0.819	0.522	0.640
TabTransformer ↑ H-Recall	0.580	0.720	**0.870**	0.570

TabTransformer achieves the **highest High Recall (0.870)** of all individual models. Custom PyTorch implementation used due to library incompatibility (x-transformers >1.x). Early-stopped at epoch 54 (TabNet).

🔬 NB-05 · META-LEARNER SELECTION

Base models: CatBoost + XGBoost + TabTransformer → 9 meta-features per woman (3 models × 3 class probabilities)

META-LEARNER	ACC.	AUC-ROC	H-RECALL
Standard LR	0.853	**0.964**	0.609
Weighted LR (3× High) Selected	0.853	**0.964**	0.652
Weighted + Boosted TT	0.853	**0.964**	0.652

Weighted LR selected: same performance as boosted variant, simpler to calibrate on real data.

🏆 FINAL STACKING ENSEMBLE — TEST SET RESULTS

90.0%	0.972	87%	88%
ACCURACY	AUC-ROC	HIGH RECALL	MACRO F1

CLASS	PRECISION	RECALL	F1	N
Low Risk	**0.97**	0.89	**0.93**	37
Moderate Risk	**0.92**	**0.91**	**0.92**	90
High Risk	0.74	**0.87**	0.80	23
Macro Avg	**0.88**	**0.89**	**0.88**	150

Zero High Risk → Low Risk misclassifications. The most dangerous triage error does not occur on the test set.

📈 NB-06 · BAYESIAN INFERENCE (PYMC NUTS)

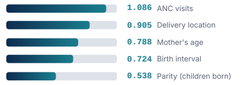
SAMPLER: **NUTS (MCMC)** · CHAINS: **2**
TUNING DRAWS: **1,000** · POSTERIOR DRAWS: **1,000**
R-HAT: **1.003** ✓ · ESS: **3,477** ✓

Output: posterior mean + 95% Highest Density Interval (HDI) per woman. Three clinical interpretability patterns observed:

DEFINITIVE	BORDERLINE	CONFIDENT LOW
P(High) = [0.97–1.00]	**P(High) = [0.30–0.86]**	**P(High) = [0.00–0.01]**
Immediate referral indicated	Clinical review recommended	Standard monitoring

🔍 NB-07 · SHAP EXPLAINABILITY (CATBOOST)

TreeSHAP global feature importance (Mean |SHAP|) — all explanations clinically coherent with WHO evidence:



Poverty (v190) and ANC attendance (m14_1) are the strongest predictors — consistent with WHO evidence on socioeconomic determinants of maternal mortality.

📱 SO3 · STREAMLIT CHW APP (PILOT)

- 13-variable input form (aligned to KDHS variables)
- Stacking ensemble prediction with risk level & probability
- Bayesian posterior mean + 95% HDI displayed per prediction
- AI-generated clinical explanation (2–3 sentences, plain language)
- Three WHO-aligned CHW recommendations per risk tier
- Automatic referral letter generation for High Risk cases
- **English / Kiswahili toggle** for CHW accessibility

AI API cost: ~$0.003 per prediction request. Free credits available for pilot validation. App run: `streamlit run 08_Pilot_App.py`

✅ KEY PILOT FINDINGS

- The stacking ensemble (CatBoost + XGBoost + TabTransformer) outperforms all individual models on AUC-ROC **(0.9717)** and macro F1 **(0.88)**
- TabTransformer, despite lower overall accuracy, captures high-risk women that tree-based models miss (Recall = **0.870**), validating its role in the ensemble
- Bayesian MCMC achieves excellent convergence (**R-hat = 1.003**, ESS = 3,477), enabling reliable uncertainty quantification for CHW decision support
- Poverty (wealth index) and inadequate ANC are the dominant predictors, consistent with global maternal health evidence
- Zero catastrophic misclassifications (High Risk predicted as Low Risk) on the test set — a critical safety property for triage tools

⚠️ LIMITATIONS & ETHICAL CONSIDERATIONS

- **Simulated data:** Pilot uses 1,000 synthetic records; generalisability to real KDHS (n=10,391) requires validation
- **Self-reported data:** KDHS responses subject to recall bias (e.g., ANC visit counts, delivery location)
- **Class imbalance:** High Risk class (15%) — SMOTE mitigates but does not eliminate imbalance effects
- **Ethics:** USIU-Africa IRB + NACOSTI approval pending; KDHS data use agreement signed; all data fully anonymised
- **CHW literacy:** Interface validated against simulated users only; real CHW usability testing planned
- **Missing variables:** v457 and v481 empty in KDHS 2022-IR release

🚀 IMPLEMENTATION PHASE — NEXT STEPS

1. Load real KDHS 2022 dataset (**10,391 women**) and retrain full pipeline
2. Optuna hyperparameter tuning — 100 trials per model (CatBoost, XGBoost, TabTransformer)
3. Calibrate risk thresholds on real population distributions
4. Bayesian inference on full dataset with geographic stratification by province (v024)
5. Deploy Streamlit app on **Streamlit Community Cloud**
6. Clinical validation with 5+ CHW participants (semi-structured interviews)
7. Final dissertation written and defended — **USIU-Africa, 2027**

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github.com/SelmahT/maternal-health-risk-kenya

Data Sources & Tools
Kenya DHS 2022-IR · DHS Program (dhsprogram.com) · Python 3.11
CatBoost · XGBoost · PyTorch · PyMC · SHAP · Streamlit · Anthropic AI API

GitHub Repository
github.com/SelmahT/maternal-health-risk-kenya